

Python Interview Questions

Day 02 • Questions, Answers & Examples

1. What is dynamic typing in Python?

Answer: Dynamic typing means that you do not need to declare a variable's data type explicitly. Python determines the type of an object at runtime, and the same variable name can refer to objects of different types at different times.

Example:

Example:

```
x = 10
print(type(x))
x = "Python"
print(type(x))
```

Output:

```
<class 'int'>
<class 'str'>
```

2. What is the difference between int, float, and str in Python?

Answer: int is used for whole numbers, float is used for numbers with decimal values, and str is used for text or a sequence of characters.

Example:

Example:

```
age = 21          # int
price = 99.5      # float
name = "Python"  # str

print(type(age))
print(type(price))
print(type(name))
```

Output:

```
<class 'int'>
<class 'float'>
<class 'str'>
```

3. What is type conversion in Python?

Answer: Type conversion is the process of changing a value from one data type to another. Common functions used for type conversion are int(), float(), str(), and bool().

Example:

Example:

```
age = "21"
age = int(age)
print(age)
print(type(age))
```

Output:

```
21
<class 'int'>
```

4. What are operators in Python?

Answer: Operators are symbols or keywords used to perform operations on values and variables. Common types include arithmetic, comparison, logical, assignment, identity, membership, and bitwise operators.

Example:

```
Example:
a = 10
b = 3

print(a + b)    # arithmetic
print(a > b)    # comparison
```

Output:

```
13
True
```

5. What is the difference between implicit and explicit type conversion?

Answer: Implicit type conversion is automatically performed by Python when it safely converts one data type to another. Explicit type conversion is manually performed by the programmer using functions such as `int()`, `float()`, or `str()`.

Example:

```
Example:

# Implicit conversion
x = 10
x = x + 2.5
print(x)

# Explicit conversion
age = "21"
age = int(age)
print(age)
```

Output:

```
12.5
21
```